Conner Jackson
Analytical Report 2
Pixelizing Societies from Space
April 17, 2026

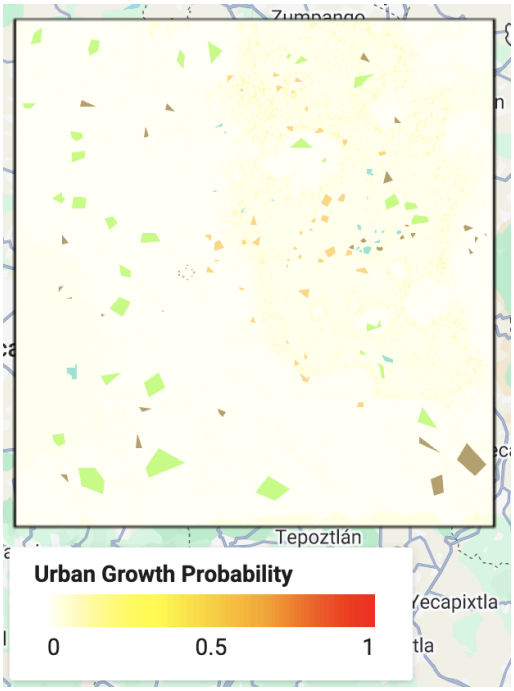
Methods

This analysis models the probability of future urban expansion using a supervised classification approach in Google Earth Engine. The dependent variable was derived from the urban change map created in Report 1, where pixels that transitioned from non-urban to urban were assigned a value of 1, and all other pixels were assigned 0. This allows the model to focus specifically on where urban growth has occurred. Three predictor variables were selected based on common patterns of urban development: distance to existing urban areas, elevation, and slope. Distance to urban areas captures the tendency for cities to expand outward from existing development, while elevation and slope represent physical constraints that can limit construction. Each variable was prepared as a spatial layer.

Distance to urban areas was calculated from the earlier classification, while elevation data was obtained from the SRTM dataset and used to derive slope. These variables were combined and sampled across the study area. A Random Forest model with 10 trees was trained on 70% of the data and tested on the remaining 30%. The model was set to output probabilities, resulting in a continuous surface representing the likelihood of future urbanization.

Prediction Map

The prediction map shows a gradient of urban expansion probability, from low to high likelihood. Higher probabilities are generally concentrated near existing urban areas and in flatter, lower-elevation regions. Lower probabilities appear in more distant or physically constrained areas. Rather than predicting exact future development, the map highlights areas where urban growth is more likely under conditions similar to those observed in the past.



Results

The model results show that distance to existing urban areas is the most important predictor, followed by elevation and slope. This reflects typical urban growth patterns, where development tends to occur near existing infrastructure and built-up areas. The spatial pattern of predicted growth seems reasonable, with higher possibilities around the edges of current urban zones. This aligns with common processes such as urban sprawl and edge expansion. Elevation and slope also play a role, highlighting that flatter and more accessible land is more likely to be developed. However, these results should be interpreted carefully. The model illustrates relationships in the data but does not fully explain the underlying causes of urban growth. For example, proximity to urban areas likely reflects access to roads, jobs, and services, which are not directly included in the model.

Limitations

A key limitation is that the model only includes a small set of physical variables and does not account for social, economic, or policy factors that strongly influence urban development. As a result, it simplifies a complex process. The model also assumes that future growth will follow the same patterns as past growth, which may not be the case if conditions change. Additionally, the binary change map does not distinguish between different types of urban development, such as dense infill versus low-density expansion. There are also potential errors carried over from the original land cover classification, which may affect the model's accuracy. Finally, the model does not account for planning constraints or protected areas that could limit development, even in high-probability locations.

Conclusion

Overall, the model provides a useful approximation of where urban expansion is most likely to occur based on past patterns and basic spatial factors. It highlights the importance of proximity to existing urban areas and terrain suitability. At the same time, the results demonstrate the limitations of relying only on spatial data. While the model produces realistic patterns, it should be seen as a simplified representation rather than a definitive prediction of future urban growth.

Additional

AI tools were used to help interpret data, develop Google Earth Engine code, and assist in structuring the written analysis.
https://code.earthengine.google.com/c354da1d28b9cb3e8311fab29f12b5d3